

Predicting Online Purchase Intention and Discovering Shopper Segments

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1 Executive Summary

This project analyzes 12,330 online shopping sessions from the UCI Online Shoppers Purchasing Intention Dataset. The first objective is to predict whether a session ends in a purchase, with purchase (**Revenue = TRUE**) treated as the positive class. The second objective is to discover behavioral shopper segments without using the purchase outcome. After data-quality checks, the observations were divided into stratified training, validation, and test sets. Logistic regression and K-nearest neighbors were developed using only the training data, while tuning decisions and the logistic probability threshold were made using validation performance. The untouched test set was used only after all choices were fixed. The final predictive model was **Logistic (threshold = 0.12)**. It achieved test accuracy of **78.4%**, sensitivity of **86.8%**, specificity of **76.9%**, and balanced accuracy of **81.8%**. For segmentation, K-means was applied only to the ten genuinely quantitative predictors after standardization. The elbow diagnostic selected **4 clusters**. These clusters differed in browsing intensity, engagement, and exit behavior, and their post-hoc purchase rates also varied. Browsing behavior therefore contains useful predictive information, although class imbalance and the assumptions of both logistic regression and K-means limit interpretation.

2 A. Data Understanding and Exploratory Analysis

2.1 A1. Import, structure, and data quality

Table 1: Dataset dimensions and basic data-quality checks.

Item	Value
Rows	12330
Columns excluding row_id	18
Missing values	0
Exact duplicate records	125

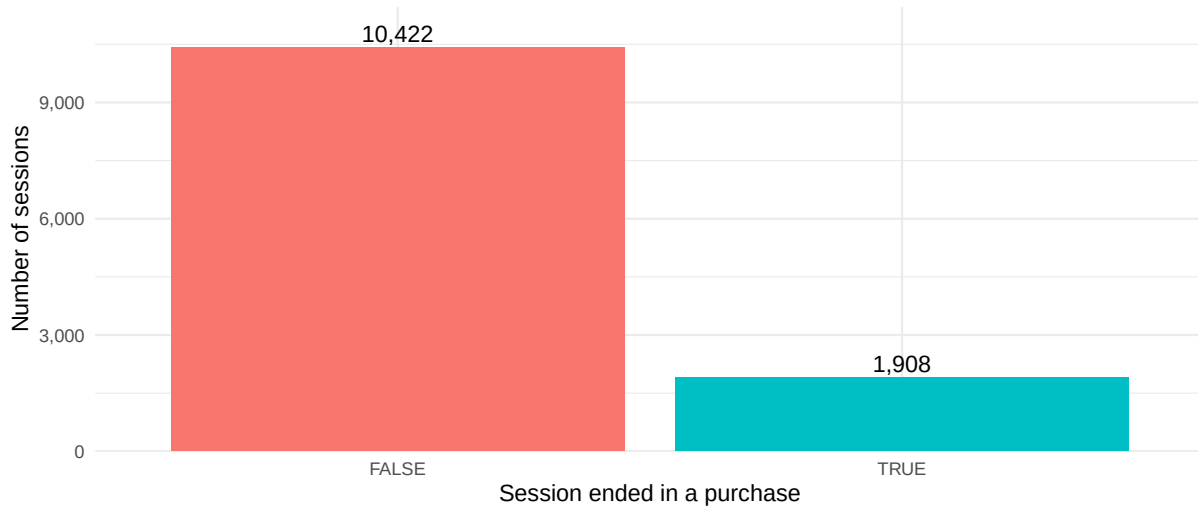
The imported dataset contains 12,330 sessions and 18 analysis variables. The ten genuinely quantitative predictors were stored as integer or numeric variables. The seven categorical predictors and binary response were converted to factors. Operating-system, browser, region, and traffic-type codes were retained as category labels rather than treated as measurements. There were 0 missing values and 125 exact duplicate records. Checks found no negative page counts or durations, no rates outside 0 to 1, no SpecialDay values outside 0 to 1, and no negative PageValues. Exact duplicates were retained because each row represents a recorded session, and identical observed values do not prove that two records describe the same event. No observations were silently deleted.

2.2 A2. Class balance and naive benchmark

Table 2: Counts and proportions of purchase outcomes.

Revenue	n	Proportion
FALSE	10422	0.8453
TRUE	1908	0.1547

Purchase outcomes are strongly imbalanced

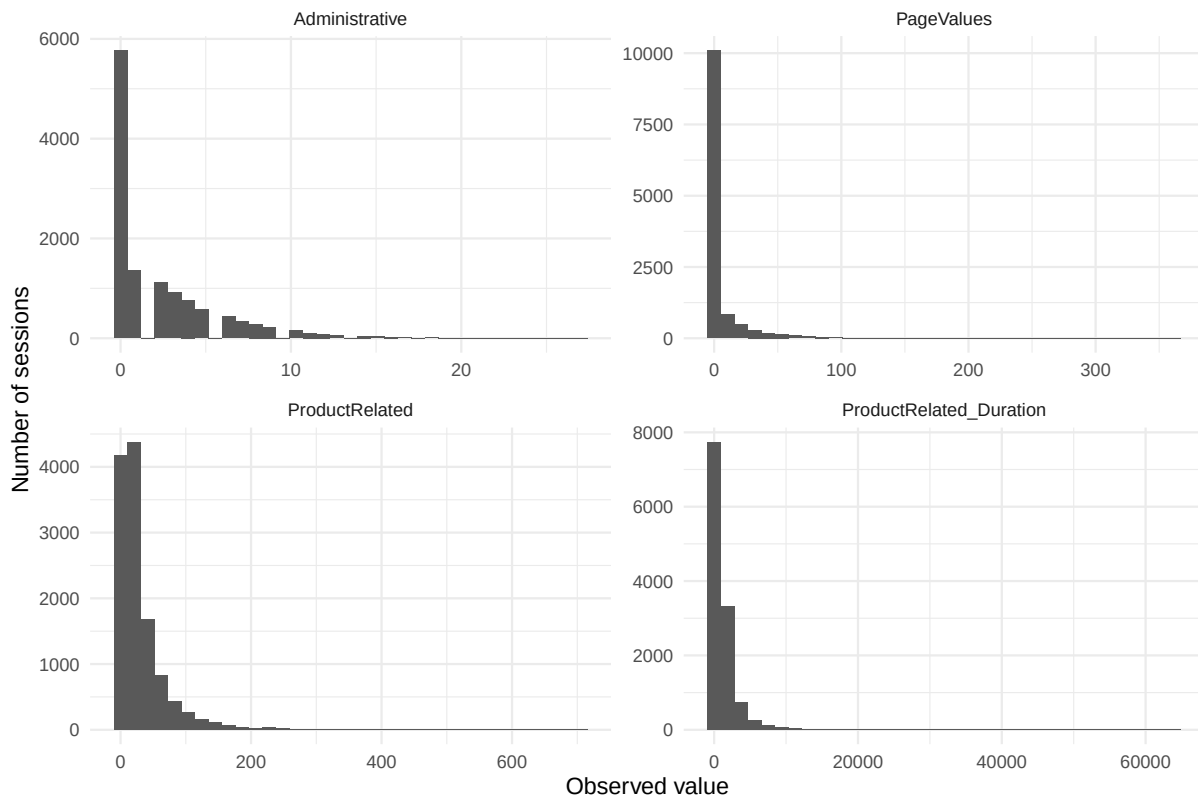


The naive majority-class classifier has accuracy of 84.5%. That accuracy is misleading because the classifier never predicts a purchase and therefore has zero sensitivity for the positive class.

2.3 A3. Exploratory visual analysis

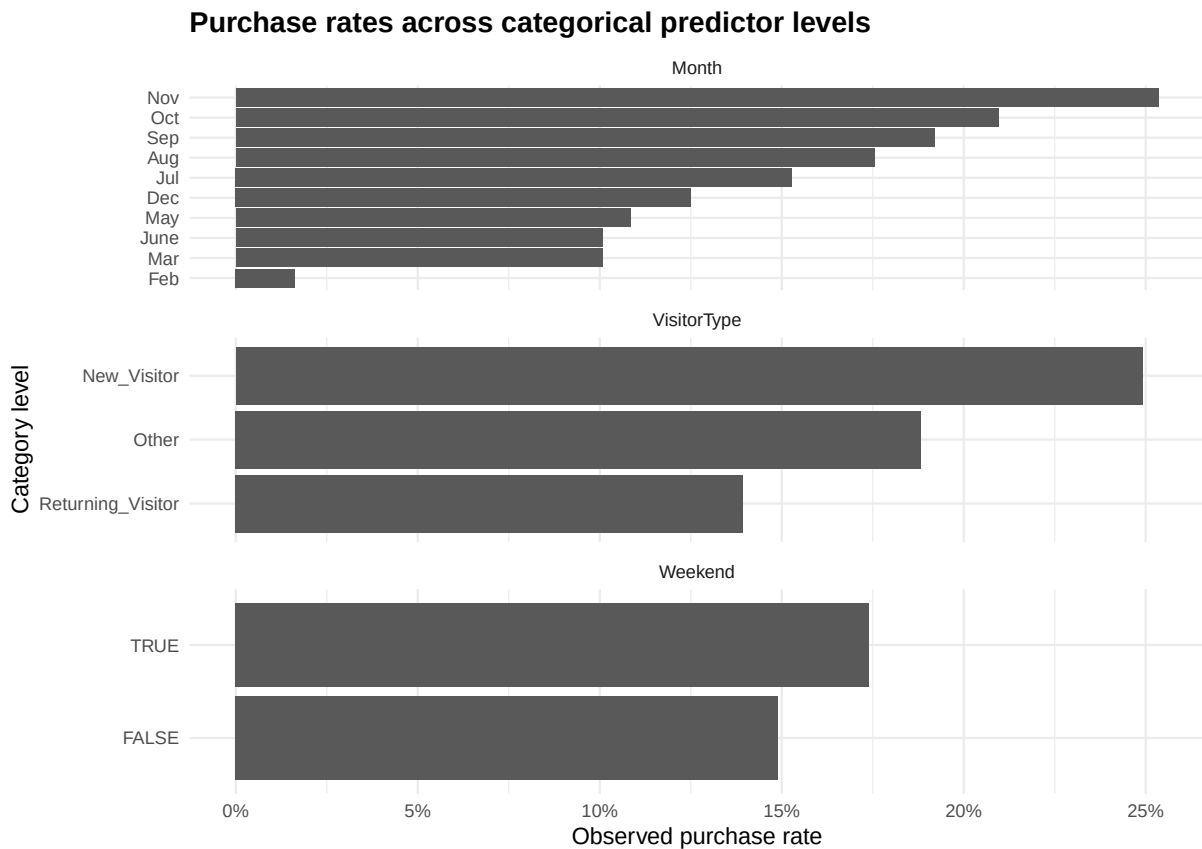
2.3.1 Distributions of quantitative variables

Distributions of selected browsing variables



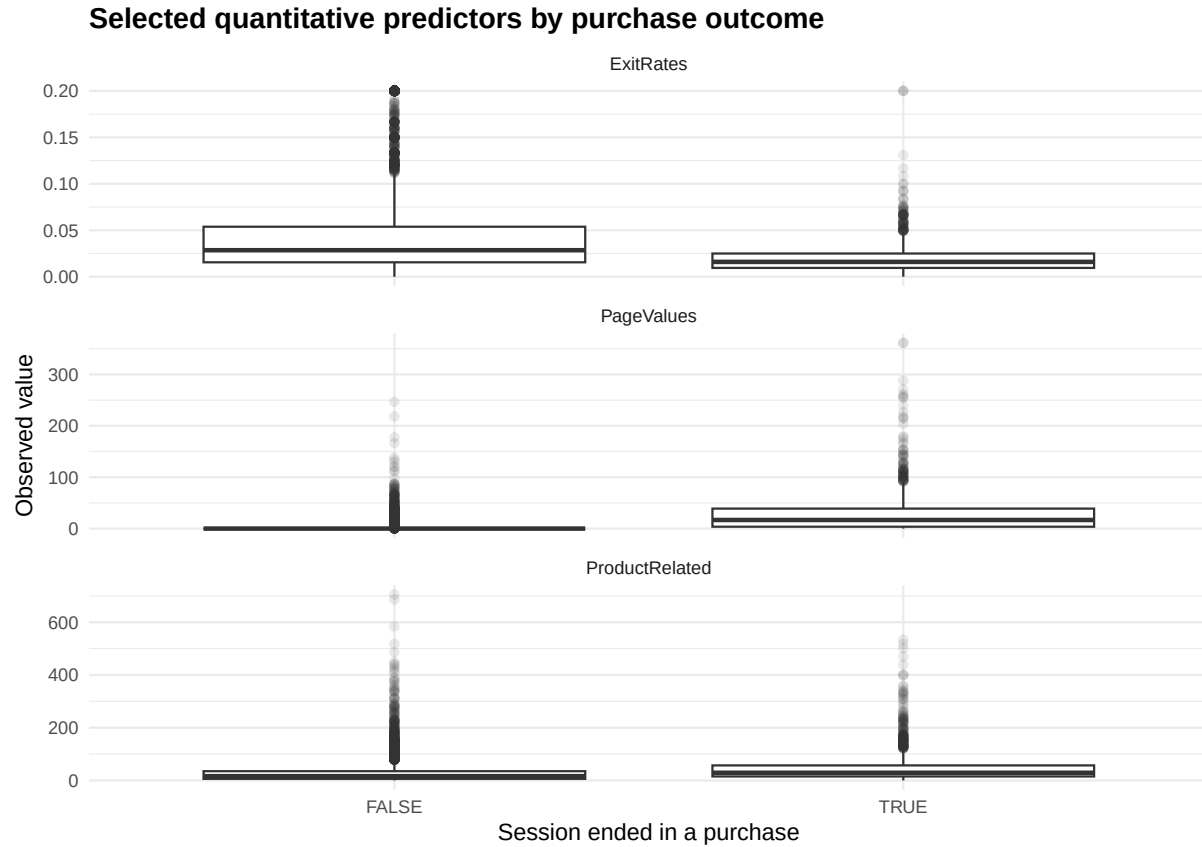
The selected page-count, duration, and page-value variables are strongly right-skewed. Most sessions show modest activity, while a small number show very high engagement.

2.3.2 Purchase rates for categorical predictors



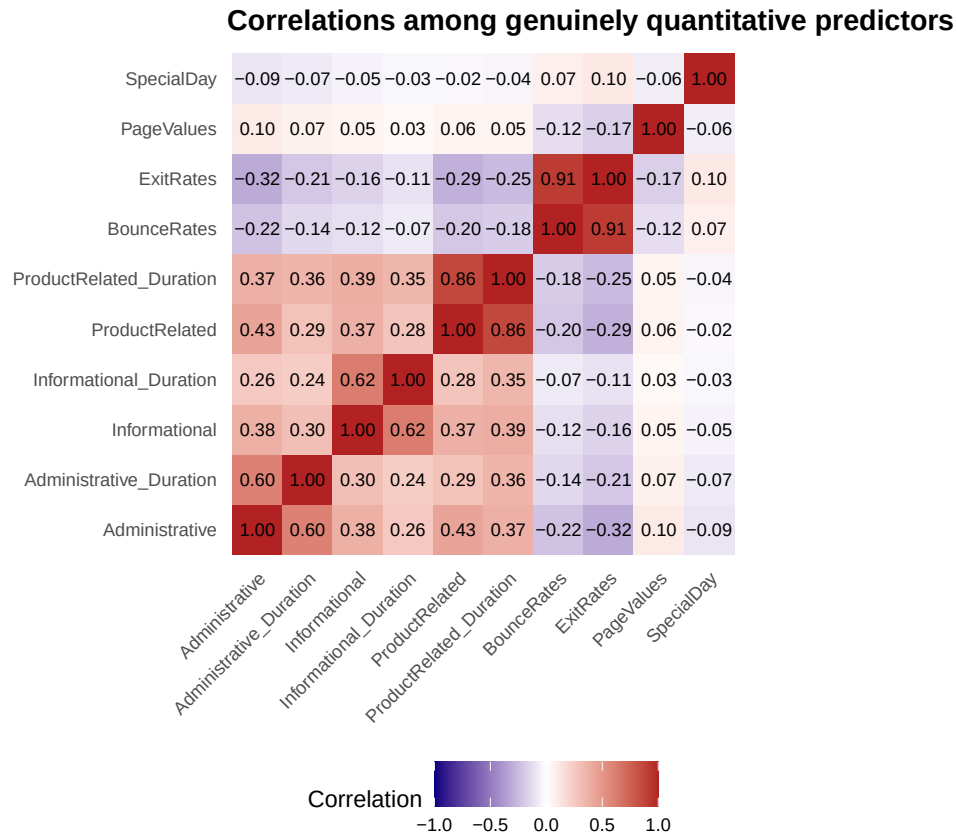
Purchase rates differ across months and visitor types, while the weekend difference is more modest. These are descriptive comparisons rather than causal effects.

2.3.3 Quantitative predictors by outcome



Purchasing sessions tend to show higher **PageValues** and more product-related browsing, while exit rates tend to be lower. Considerable overlap remains between the two outcome groups.

2.3.4 Correlation display



Page counts are positively associated with their corresponding duration measures. **BounceRates** and **ExitRates** are also positively related. These relationships may make individual logistic coefficients less stable.

3 B. Data Splitting and Preprocessing

3.1 B1. Training, validation, and test sets

The assignment requires an approximately 70% training and 30% holdout split while also requiring separate validation and test evaluations. The 30% holdout portion was therefore divided evenly, producing an approximately 70/15/15 stratified split.

Table 3: Dimensions and purchase proportions for each split.

Set	Rows	PurchaseRate
Training	8630	0.1547
Validation	1849	0.1547
Test	1851	0.1551

Table 4: Verification that datasets do not overlap.

Pair	SharedRows
Training vs validation	0
Training vs test	0

Pair	SharedRows
Validation vs test	0

Purchase proportions are nearly identical across the three datasets, and all pairwise overlap counts are zero.

3.2 B2. Method-specific preprocessing

Logistic regression uses meaningful factors and the reference levels below.

Table 5: Reference levels for categorical predictors.

Predictor	Baseline
Month	Nov
OperatingSystems	2
Browser	2
Region	1
TrafficType	2
VisitorType	Returning_Visitor
Weekend	FALSE

KNN uses a one-hot numeric design matrix. Centering and scaling constants were estimated only from the training data and then applied to validation and test data with the same column structure. Coded values such as browser and region cannot be treated as ordinary numbers because their numerical differences do not represent meaningful behavioral distances.

4 C. Logistic Regression

4.1 C1. Main-effects model

The fitted model is:

Revenue ~ Administrative + Administrative_Duration + Informational + Informational_Duration + ProductRelated + ProductRelated_Duration + BounceRates + ExitRates + PageValues + SpecialDay + Month + OperatingSystems + Browser + Region + TrafficType + VisitorType + Weekend

Purchase (Revenue = TRUE) is the positive class.

Table 6: Selected logistic-regression odds ratios.

Term	OR	CI low	CI high	p-value
PageValues	1.087	1.081	1.093	0.000
MonthMay	0.343	0.268	0.438	0.000
MonthDec	0.327	0.252	0.422	0.000
MonthMar	0.351	0.271	0.451	0.000
ExitRates	0.000	0.000	0.000	0.000
TrafficType13	0.447	0.271	0.707	0.001
TrafficType3	0.655	0.504	0.846	0.001
MonthJune	0.416	0.227	0.718	0.003
MonthSep	0.554	0.371	0.813	0.003
Browser12	13.824	2.184	79.933	0.003

4.2 C2. Coefficient interpretations

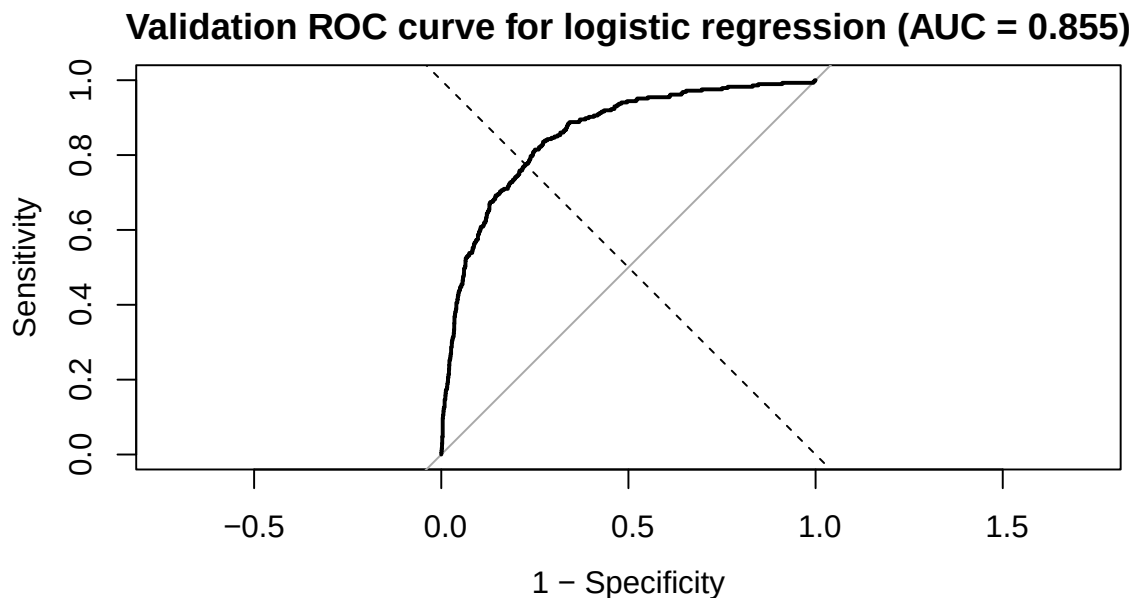
Holding all other included predictors constant, an additional 100 seconds on product-related pages multiplies the estimated purchase odds by **1.006**.

Holding all other included predictors constant, a weekend session has estimated purchase odds **1.141 times** those of an otherwise comparable weekday session.

Holding all other included predictors constant, a new visitor has estimated purchase odds **1.163 times** those of an otherwise comparable returning visitor.

These are conditional associations, not causal effects.

4.3 C3. Validation ROC curve, threshold, and confusion metrics



The validation AUC is 0.855. The ROC threshold closest to the top-left corner is 0.12.

Table 7: Validation metrics for two logistic-regression thresholds.

Threshold	Cutoff	Accuracy	Sensitivity	Specificity	Precision	Error rate	Balanced acc.
0.50	0.50	0.868	0.301	0.972	0.662	0.132	0.636
ROC closest to top-left	0.12	0.760	0.815	0.750	0.373	0.240	0.782

Table 8: Logistic validation confusion matrix at threshold 0.50.
Rows are predicted classes and columns are actual classes.

Predicted	FALSE	TRUE
FALSE	1519	200
TRUE	44	86

Table 9: Logistic validation confusion matrix at the ROC-selected threshold. Rows are predicted classes and columns are actual classes.

Predicted	FALSE	TRUE
FALSE	1172	53
TRUE	391	233

For a business use case in which missing a likely purchaser is more costly than showing an unnecessary promotion to a non-purchaser, the ROC-selected threshold is preferred because it provides a better balance between sensitivity and specificity than the default threshold.

4.4 C4. Logistic-model limitations

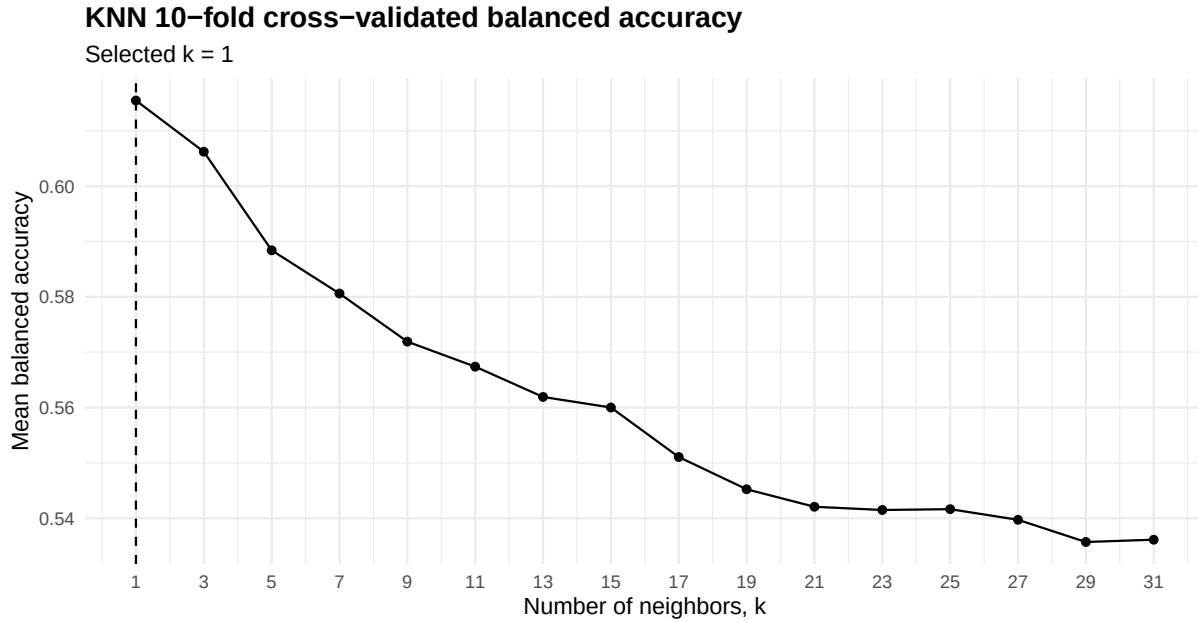
First, a main-effects logistic model assumes that each quantitative predictor has a linear effect on log-odds and does not automatically represent nonlinear relationships or interactions. Second, the strong class imbalance can make overall accuracy look favorable even when purchase detection is weak. Correlated browsing measures can also make individual coefficients unstable.

5 D. K-Nearest Neighbors and 10-Fold Cross-Validation

5.1 D1. Selecting the number of neighbors

Table 10: Ten-fold cross-validation results for odd values of k.

k	misclassification_rate	sensitivity	specificity	balanced_accuracy
1	0.183	0.324	0.907	0.615
3	0.153	0.258	0.955	0.606
5	0.148	0.207	0.970	0.588
7	0.145	0.183	0.978	0.581
9	0.143	0.159	0.985	0.572
11	0.142	0.146	0.989	0.567
13	0.142	0.133	0.991	0.562
15	0.142	0.129	0.991	0.560
17	0.144	0.110	0.992	0.551
19	0.145	0.097	0.993	0.545
21	0.146	0.091	0.994	0.542
23	0.146	0.088	0.995	0.541
25	0.145	0.088	0.995	0.542
27	0.145	0.084	0.996	0.540
29	0.147	0.076	0.995	0.536
31	0.146	0.076	0.996	0.536



The selected value is 1, which had the highest cross-validated balanced accuracy. A value was considered a near-tie if it was within 0.005 of the best balanced accuracy. No larger value of k met that rule, so the near-tie rule did not change the selection.

5.2 D2. Validation performance and model comparison

Table 11: KNN validation confusion matrix. Rows are predicted classes and columns are actual classes.

Predicted	FALSE	TRUE
FALSE	1418	198
TRUE	145	88

Table 12: Validation comparison of the selected KNN and logistic classifiers.

Model	Accuracy	Sensitivity	Specificity	Precision	Error rate	Balanced acc.
Logistic (threshold = 0.12)	0.760	0.815	0.750	0.373	0.240	0.782
KNN (k = 1)	0.814	0.308	0.907	0.378	0.186	0.607

Based on validation balanced accuracy, sensitivity, and accuracy, the selected final model is **Logistic (threshold = 0.12)**.

5.3 D3. Role of scaling and k

Standardization is essential because KNN is distance-based. Without scaling, variables with large numerical ranges, such as duration, would dominate variables with small ranges, such as bounce rate. Very small values of k produce flexible boundaries with low bias but high variance. Very large values produce smoother boundaries with higher bias and lower variance. The cross-validation curve shows where this tradeoff gives the strongest out-of-sample balanced accuracy.

6 Final Predictive Evaluation on the Untouched Test Set

The test set was not used for preprocessing estimates, threshold selection, KNN tuning, or model selection.

Table 13: Untouched test-set metrics for Logistic (threshold = 0.12)

TN	FN	FP	TP	accuracy	sensitivity	specificity	precision	misclassification_rate	balanced_accuracy
1203	38	361	249	0.784	0.868	0.769	0.408	0.216	0.818

Table 14: Final test confusion matrix. Rows are predicted classes and columns are actual classes.

Predicted	FALSE	TRUE
FALSE	1203	38
TRUE	361	249

The final model achieved test accuracy of 78.4%, sensitivity of 86.8%, specificity of 76.9%, precision of 40.8%, and balanced accuracy of 81.8%.

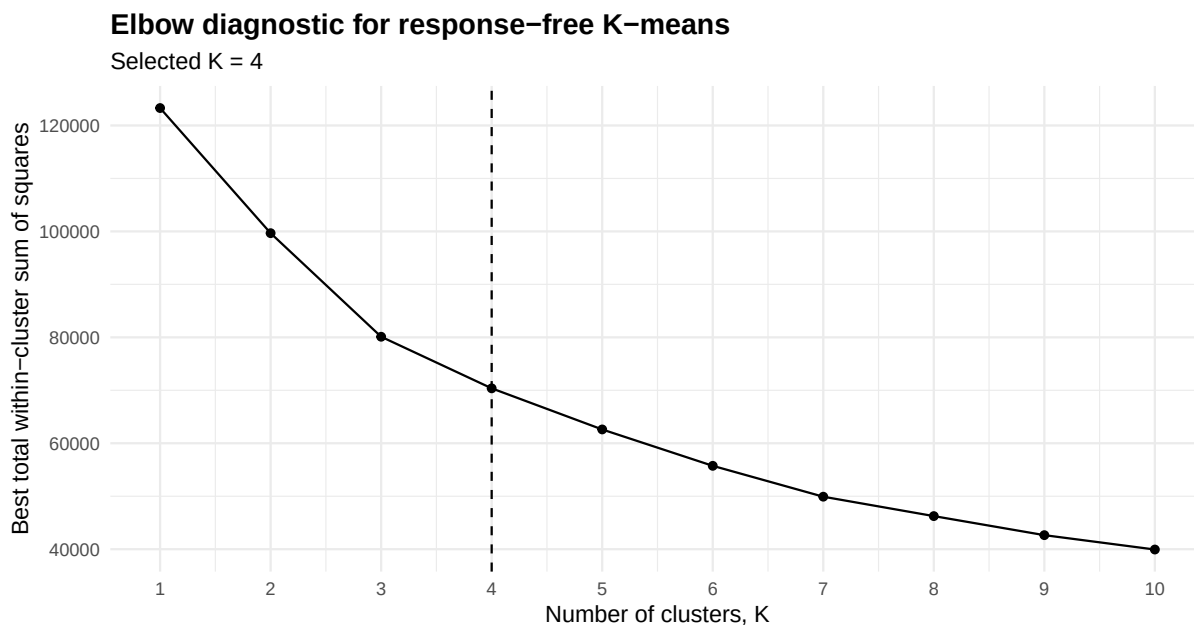
Because logistic regression was selected, its untouched test-set AUC is 0.91.

7 E. Response-Free K-Means Clustering

7.1 E1. Clustering feature matrix

Before any clustering choice was made, **Revenue** was removed. The clustering matrix contains only the ten genuinely quantitative predictors. Integer-coded categorical variables were excluded because their codes do not define meaningful Euclidean distances. All ten variables were standardized.

7.2 E2. Selecting the number of clusters



The final value is 4. It was selected as the point with the greatest perpendicular distance from the straight line joining the first and last points of the standardized elbow curve. This identifies the strongest bend, after which additional clusters provide smaller reductions in within-cluster variation.

7.3 E3. Final clustering and behavioral profiles

Table 15: Cluster sizes. Total within-cluster variation = 70371.6

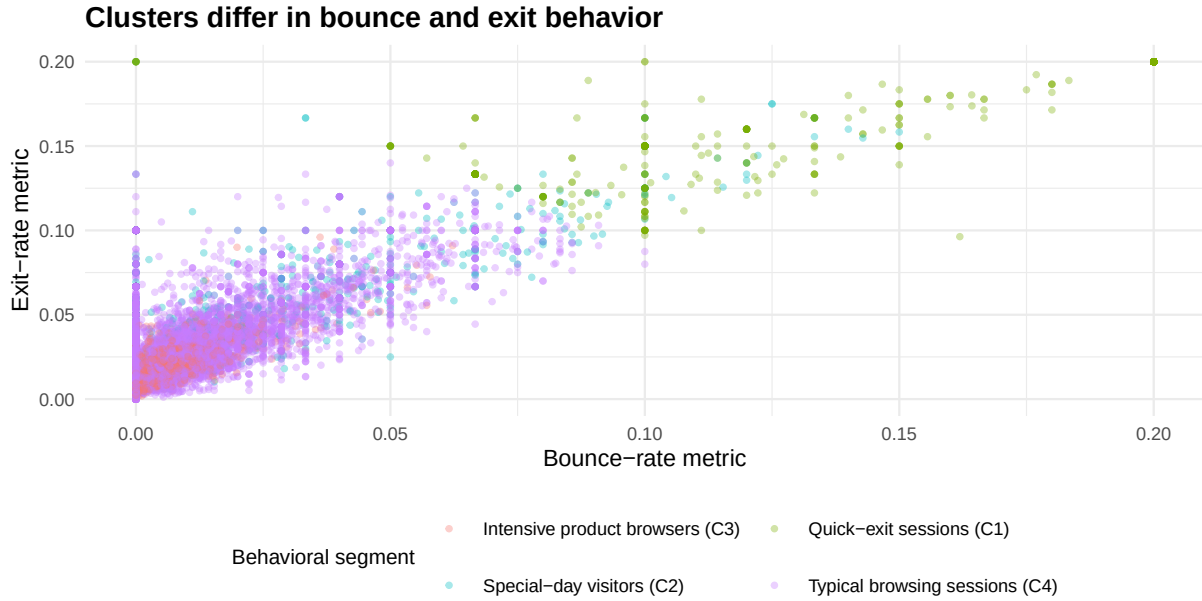
Cluster	Size	BehavioralName
1	974	Quick-exit sessions (C1)
2	941	Special-day visitors (C2)
3	1499	Intensive product browsers (C3)
4	8916	Typical browsing sessions (C4)

Table 16: Cluster profiles: mean standardized values (z-scores).

Variable	C1	C2	C3	C4
Administrative	-0.68	-0.38	1.54	-0.15
Administrative_Duration	-0.45	-0.29	1.32	-0.14
Informational	-0.39	-0.23	1.67	-0.22
Informational_Duration	-0.24	-0.18	1.36	-0.18
ProductRelated	-0.66	-0.11	1.53	-0.17
ProductRelated_Duration	-0.60	-0.15	1.44	-0.16
BounceRates	3.10	-0.01	-0.33	-0.28
ExitRates	2.90	0.13	-0.49	-0.25
PageValues	-0.32	-0.21	0.21	0.02
SpecialDay	0.00	3.13	-0.23	-0.29

The cluster names are based only on their browsing profiles, not on purchase outcome. Positive z-scores indicate above-average values and negative z-scores indicate below-average values. C1 is dominated by high bounce and exit rates, C2 by high SpecialDay values, C3 by high page counts and durations, and C4 is the large group closest to typical browsing behavior.





The first plot emphasizes differences in product browsing intensity. The second shows how the segments differ in low-engagement and quick-exit behavior.

7.4 E4. Post-hoc purchase interpretation and limitations

Only after the final cluster assignments were fixed was **Revenue** reattached.

Table 17: Post-hoc purchase rates by response-free cluster.

Cluster	Sessions	PurchaseRate	BehavioralName
1	974	0.006	Quick-exit sessions (C1)
2	941	0.061	Special-day visitors (C2)
3	1499	0.286	Intensive product browsers (C3)
4	8916	0.159	Typical browsing sessions (C4)

The post-hoc rates show whether behavioral segments also happen to differ in purchase behavior. These differences do not mean that purchase outcome created the clusters.

K-means has several limitations here. It relies on Euclidean distance and tends to favor roughly spherical clusters. It is sensitive to skewness and outliers, both of which are present in the browsing variables. It also excludes useful categorical information, can vary with initialization, and does not provide a uniquely correct value of K .

8 F. Conclusions, Communication, and Reproducibility

The analysis kept the test set untouched until the final model was chosen and used sensitivity and balanced accuracy because purchases were much less common than non-purchases. Logistic regression performed better than KNN on validation balanced accuracy and was selected as the final classifier. K-means found four browsing groups with different engagement patterns. These groups are useful descriptions of the data, but they should not be treated as fixed or causal customer types.

All random operations use a fixed seed of 380. The source imports the dataset, performs data preparation, reproduces every table and figure, and can be run from a clean R session after the required packages are installed.

9 References

Sakar, C., & Kastro, Y. (2018). *Online Shoppers Purchasing Intention Dataset*. UCI Machine Learning Repository. <https://doi.org/10.24432/C5F88Q>